



Seeking for AI

Realistic Datasets for Photons and Neutrons AI methodologies

McStas —
neutrons

McXtrace — X-
rays



PaNRAID School · supported by PEPR DIADEM



This school's objective

Explore how AI can leverage simulated photon and neutron datasets to:

1. Solve inverse problems (measurement $\rightarrow$ parameters)
2. Optimize experiments (find best parameters)
3. Train surrogate models (parameters $\rightarrow$ measurement)

For all this, we encourage you to use AI chats/agents.



Why "Photons and Neutrons"?

Shared X/n Physics and Simulation Capabilities

Similar Interaction Processes:

- **Absorption:** Energy transfer to the material (e.g., photoelectric effect for X-rays, neutron capture).
- **Scattering:**
 - ▶ *Elastic:* Structure analysis (e.g., Bragg diffraction for X-rays, neutron diffraction).
 - ▶ *Inelastic:* Dynamics analysis (e.g., Raman scattering for X-rays, neutron inelastic scattering).
- **Decay:** Time-resolved studies (e.g., fluorescence, neutron activation).



Why "Photons and Neutrons"?

The Real World:

- **Parameters** → **[Instrument]** → **"Measurement"**

Simulation Tools:

- **McStas**: Monte Carlo simulator for neutron scattering instruments.
- **McXtrace**: Monte Carlo simulator for X-ray instruments.
- Both enable realistic modeling of beamlines, detectors, and sample interactions.

Why Simulations Matter:

- **Realism**: Capture complex instrument geometries and material responses.
- **Flexibility**: Test hypothetical setups.
- **Cost-Effective**: Avoid expensive trial-and-error on real instruments/BL.



Why Realistic AI Datasets?

The Problem with Experimental Data

- **Lack of Annotations:**
 - Raw data often lacks metadata (e.g., sample composition, instrument settings).
 - Manual annotation is time-consuming and error-prone.
- **Publications as Implicit Annotations:**
 - Publications describe experimental conditions, but formats vary widely.
 - Not machine-readable or standardized for AI training.



Why Realistic AI Datasets?

The Power of Simulated Data

- **Controlled Parameters:**
 - Every input (e.g., sample thickness, detector angle) is known and adjustable.
 - Outputs (e.g., scattering patterns, spectra) are directly linked to inputs.
- **Pipeline Workflow:**
 - **Parameters** → **[Simulation]** → **Virtual “Measurement”**
- Example: Simulate an experiment with varying incident energy.



General AI Methodology

- **AI as a Universal Function Approximator:**
- **Black Box Concept:**

$X = \text{Input} \rightarrow [\text{AI Model}] \rightarrow Y = \text{Output}$

- **Universal Approximation Theorem:** A neural network can approximate any continuous function given sufficient data and complexity.
- **General Behavior:**
 - All AI methods (CNNs, RNNs, SVMs) follow the same principle: map inputs to outputs via learned weights.
 - Differences lie in architecture, training efficiency, and interpretability.



General AI Methodology

Parameters → [Simulation] → Virtual “Measurement”

X → [AI] → Y

	Input X=Parameters	Input X=Measurements
Output Y=Parameters	Optimisation	Inverse problem : classification, regression
Output Y=Measurements	Simulation, Surrogate	De-noising, de-convolution, segmentation



AI algorithm types

Inverse Problem:

- **Goal:** Reconstruct input parameters from measured data.
- **AI Approach:**

$X = \text{Measurement} \rightarrow [\text{AI Algorithm}] \rightarrow Y = \text{Parameters}$

- The inferred *Parameters* can be discrete (class) or continuous (regression).
- Example: Predict sample composition from a scattering pattern.



AI algorithm types

Surrogate Models:

- **Purpose:** Replace slow simulations with fast AI models.
- **Pipeline:**

$X = \text{Parameters} \rightarrow [\text{AI Model}] \rightarrow Y = \text{"Measurement"} \text{ (faster)}$

- **Use Case:** Fast phase-space exploration, Bayesian analysis.



AI algorithm types

Model optimization:

- **Purpose:** Find best parameters to get min/max signal.
- **Pipeline:**

$X = \text{Parameters} \rightarrow [\text{Optimizer}] \rightarrow Y = \text{Best Parameters}$

- **Use Case:** Optimization of beamline configurations.



AI algorithm types

Data Processing:

- **Goal:** process measured data.
- **AI Approach:**

$X = \text{Measurement} \rightarrow [\text{AI Algorithm}] \rightarrow Y = \text{Measurement}$

- Example: filter noise, super-resolution, segmentation.



General AI Methodology

- **Training: The Core Process**
- **Objective:** Minimize the loss function (e.g., Mean Squared Error, Cross-Entropy).
 - Loss Function: Measures the difference between predicted and actual outputs.
- **Steps:**
 - Forward Pass: Compute predictions for a batch of training data.
 - Loss Calculation: Evaluate how far predictions are from true values.
 - Backpropagation: Compute gradients of the loss with respect to weights.
 - Optimization: Update weights using an optimizer (e.g., Adam, SGD).



General AI Methodology

- **Training: The Core Process**
- **Key Factors for Success:**
 - Data Quality: High-fidelity simulations with diverse parameters.
 - Model Architecture: Match complexity to the problem (e.g., CNNs for images, Transformers for sequences, trees for classification).
 - Hyperparameters: Learning rate, batch size, number of epochs.



What You Will Learn and Practice

- **Hands-on skills**
- **Setting Up Simulations:**
 - Configure beamlines/instruments in McStas (neutrons) and McXtrace (X-rays).
 - Define sample properties (e.g., crystal structure, composition).
- **Characterizing Models:**
 - Check geometry (e.g., detector positions, slit sizes).
 - Extract parameters (e.g., resolution, flux) and results (e.g., spectra, diffraction patterns).



What You Will Use

- **Your Laptop**
- **High Performance Computing (MesoNET)**
- **Remote Computing Service (DARTS@SOLEIL)**
 - <https://data-analysis.synchrotron-soleil.fr> (herculesd/Synchrotron-SØLEIL)
- **Your Brain**

Thank you

Questions?

mcstas.org

mcxtrace.org



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